

COURSE OUTLINE

Course Title:	Implementation & Management of Corporate Cybersecurity Measures	Course Number:	420-4C2-DW
Course Hours:	60	Ponderation:	1 – 3 – 3
Credits:	2 1/3	Corequisite:	None
Prerequisite:	All courses in the program	Term:	4
Group:	1	Contact Details:	agrozdanov@dawsoncollege.qc.ca
Teacher Name:	Asen Grozdanov		

COURSE COMPETENCY STATEMENT: EZ08 - Implement and manage cybersecurity measures in companies.
This course will address competency EZ08 in its entirety.
(See page 7 for full course competency).

COURSE DESCRIPTION: This course provides a comprehensive approach to implementing and managing cybersecurity measures in corporate environments. Students will learn to carry out risk analysis, identify and prioritize information assets, and assess threats and vulnerabilities. The curriculum covers securing information through the application of norms, policies, and standards, correcting high-risk vulnerabilities, implementing access controls, and validating security measures in a simulated environment. Students will also develop skills in managing incidents—from detection through resolution and reporting—and in raising user awareness through effective communication, case studies, and recommendations for safe behaviour. Through a combination of theoretical instruction and hands-on labs on realistic infrastructure, participants will gain practical experience in implementing cybersecurity measures aligned with industry best practices.

COURSE ASSESSMENT: Students will be evaluated through both formative and summative assessments through a variety of means including:

- Lab exercises
- Case study analysis
- Group discussions

- Hands-on projects
- Security implementation simulations and awareness campaigns

Appropriate mastery of the course objectives is essential to Succeed both in the program and [to successfully transition to a role in industry](#).

Assessment of student performance will involve the ability to Understand and apply key Cybersecurity concepts. The passing grade for all assessments is 60%.

COURSE LEARNING OUTCOMES:

By the end of the course, students will be able to :

- Carry out risk analysis.
- Secure information.
- Manage incidents.
- Raise user awareness.

INTEGRATION:

This is a capstone course in the program. It integrates knowledge and skills from all previous courses in the Cybersecurity: Prevention and Intervention program. [Students must pass all other courses in the program to register in this course.](#)

WEEKLY SCHEDULE OF LEARNING ACTIVITIES *:

Week	Dates	Class 1	Topic	Class 2	Topic
1	Sat 30 May	3.0 hrs	Introduction to the Capstone & Risk-Driven Program Design	4.0 hrs	Lab Foundation — OPNsense Perimeter & VLAN Segmentation
2	Sun 31 May Mon 01 Jun	3.0 hrs	Information Asset Identification, Inventory & CIA Classification	4.0 hrs	Threat Modelling (STRIDE), Baseline Reconnaissance & Monitoring Deployment
3	Thu 04 Jun Fri 05 Jun	4.0 hrs	Stage 1 Submission Review & Perimeter Hardening	4.0 hrs	Linux Hardening — CIS Benchmark, SSH & Fail2Ban

4	Sat 06 Jun. Mon 08 Jun	4.0 hrs	Windows & Active Directory Hardening	4.0 hrs	Vulnerability Management — OpenVAS, CVSS Triage & Compliance Scanning
5	Thu 11 Jun. Sat 13 Jun	4.0 hrs	Stage 2 Review, Compliance Framework Mapping & Policy Drafts	4.0 hrs	Penetration Test Pre-engagement — Rules of Engagement, OPSEC & Reconnaissance
6	Mon 15 Jun. Thu 18 Jun	4.0 hrs	Penetration Test Exploitation — Network Service, Web App & Active Directory	4.0 hrs	Stage 3 Review, IDS Evasion & Pentest Report Write-up
7	Fri 19 Jun. Sat 20 Jun	4.0 hrs	Incident Response Plan v1.0, Playbooks & Wazuh Alert Tuning	5.0 hrs	Full IH&R Simulation, Digital Forensics & Awareness Campaign Build
8	Tue 23 Jun. Fri 26 Jun	4.0 hrs	Stage 3 & Final Project Document & Presentation Submission	4.0 hrs	Final Written Exam — All Seven Integrated Courses

**Learning activities are subject to change.*

Evaluation:

Component	Tentative Date	Value of element
<i>Each Project' Stage' Submission</i>	June 4, June 11, June 18, June 23	28%
<i>Presentation</i>	June 26	12%
Final Project Delivery	June 26	30%
Final Assessment Summative Assessment	June 26	30%
		100%

IMPORTANT INFORMATION REGARDING EVALUATION:

Summative Assessment:

Students must successfully pass the summative assessment element in the course. Students who are unsuccessful on the Summative Assessment will fail the course.

Late Submission Policy:

Penalties for late submissions will be as follows: 10% for each day late with no submission accepted 7 days after due date.

Missed Tests / Quizzes: *[Adjust as needed for your course]*

Students with a valid reason for missing assessments can be provided with make-up exams. For absences and illnesses of 5 days or less, a medical certificate is not required. Absences of more than 5 days require the submission of a valid medical certificate to attest that the student was not able to attend the test component on the given date and time.

Note: A minimum grade total of 60% must be achieved to successfully complete this course with a passing grade.

Literacy policy: In graded activities, teachers may deduct up to 10% for grammar, spelling, punctuation and/or syntax errors.

The Computer Science Department recognizes that literacy in all its forms (read, written, spoken) is essential to students in their careers.

Teachers may choose to incorporate a literacy component into the marking scheme for any piece of work. Teachers may use their discretion to insist that any piece of work submitted for credit is revised by the student if it is unsatisfactory with regard to literacy.

Teachers will inform all students in their courses of this policy at the beginning of each semester either by including it in the course description or otherwise.

Required/Recommended Reading

Students must create a free, online account through the *Bibliothèque et Archives Nationale du Québec (BANQ)*: <https://cap.banq.qc.ca/inscription?locale=en>

Students can access resources through the *BANQ* such as:

- O'Reilly Publishing
- LinkedIn Learning

The Institutional Student Evaluation Policy (ISEP) is designed to promote equitable and effective evaluation of student learning and is therefore an essential policy to read and understand. The policy describes the rights and obligations of students, faculty, departments, programs, and the College administration regarding evaluation in all your courses, including grade reviews and resolution of academic grievance. ISEP is available on the Dawson website (see link above). This course outline has been prepared in conformity with the Institutional Student Evaluation Policy (ISEP). The full policy is available on the Dawson College [webpage](#).

Below are excerpts of important policies and procedures outlined in ISEP:

Literacy Standards

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Teachers may choose to incorporate a literacy component into the marking scheme for any piece of work. Teachers may use their discretion to insist that any piece of work submitted for credit is revised by the student if it is unsatisfactory regarding literacy.

Teachers will inform all students in their courses of this policy at the beginning of each semester either by including it in the course description or otherwise.

Academic Integrity Policy

Cheating and Plagiarism are serious offences and will result in failure in the assignment, test or exam, or **entire evaluation component** and **may also result in failure of the course**. Further disciplinary action might be taken which may result in suspension or expulsion from the program and the College.

Every instance of cheating or plagiarism leading to a resolution that impacts a student's grade must be reported, with explanation, in writing, to the appropriate Dean. (ISEP Section V-A)

Students are asked to familiarize themselves with ISEP p. 21 to 24. In addition, the Business Administration Department enforces the following rules: Electronic dictionaries and cell phones are prohibited during tests.

Code of Conduct

"Everyone has the right to a safe and non-violent environment. Students are obliged to conduct themselves as stated in the Student Code of Conduct and in the ISEP section on the roles and responsibilities of students." (ISEP section II-D).

The full text of the Code of Conduct Policy is available on the Dawson College [webpage](#).

Your attention is directed to the ISEP provision requiring respectful behaviour and general decorum. Violation of these provisions may lead to the exclusion from the classroom and the case referred to the Director of Student Services or the Dean of Continuing Education.

Professional Conduct

Faculty members in the program are responsible for assessing student behaviour in terms of suitability to the profession, advising students that exhibit inappropriate behaviour, and reporting said behaviour to the Program Coordinator when necessary. (ISEP section IV-O.2)

Attendance Policy:

"Students should refer to the Institutional Student Evaluation Policy (ISEP section IV-C) regarding attendance."

Policy on Religious Observances Statements:

“Students observing religious holidays must inform their teachers, in writing, as prescribed in the ISEP Policy on Religious Observances, no later than the end of the second week of the impacted semester or term. This applies both to the semester or term, as well as to any final examination period.” (ISEP section IV-D).

Objective**Standard**

Statement of the Competency	Achievement Context
Implement and manage cybersecurity measures in companies.	• On realistic infrastructure. • Based on a clear mandate. • In response to simulated incidents. • Based on norms, policies, and standards. • Using public and private sources of information. • With the help of: • Tools, equipment, and software; • Reference frameworks; • Supplier documentation or scientific articles; • •
Performance Criteria for the Competency as a Whole	
• Demonstration of autonomy. • Effective priority management. • Methodical approach to work. • Detailed documentation of operations carried out. • Effective communication with stakeholders.	

Elements of the Competency	Performance Criteria
1. Develop a cybersecurity incident response plan.	1.1 Identification of relevant sources of information. 1.2 Methodical collection and analysis of information. 1.3 Prioritization of information assets. 1.4 Identification of the main threats and vulnerabilities.
2. Secure information.	2.1 Correct application of norms, policies and standards. 2.2 Effective correction of high-risk security vulnerabilities. 2.3 Application of the relevant access control measures. 2.4 Methodical validation of the effectiveness of the measures put in place. 2.5
3. Manage incidents.	3.1 Implementation of effective incident detection measures. 3.2 Rapid detection of incidents.

	3.3	Complete resolution of
	3.4	Drawing up a detailed incident report.
4. Raise user awareness.	4.1	Identification of relevant threats.
	4.2	Demonstration of the impact of different vulnerabilities.
	<u>4.3</u>	<u>Presentation of relevant case studies.</u>
	<u>4.4</u>	<u>Appropriate use of visual aids.</u>
	<u>4.5</u>	<u>Recommendations for safe behaviour.</u>